

ANATOMY OF LL BY SOCRATES – LECTURE 1

These notes present a **clear, structured, and exam-oriented comparison of the Upper Limb (UL) and Lower Limb (LL)**, followed by the **functional, clinical, and pelvic girdle anatomy** relevant for anatomy examinations.

I. COMPARISON OF UPPER LIMB (UL) AND LOWER LIMB (LL)

1. GENERAL FEATURES & DEVELOPMENT

- Both UL and LL are **appendages**.
- Limb buds develop around **day 28 of intrauterine life**.
 - UL primordium appears earlier ($\approx$ **day 26**).
 - LL primordium appears shortly afterward.

TYPICAL FEATURE OF LIMB FORMATION – ROTATION

- Limb rotation is a key developmental event that explains adult anatomy.

Upper Limb (UL)

- Rotates **laterally by 90°**.
- Flexor compartment comes to lie **anteriorly**.

Lower Limb (LL)

- Rotates **medially by 90°** (described as latero-posteromedial rotation).
- Flexor compartment comes to lie **posteriorly**.

➡ This explains why:

- Flexors are anterior in UL.
- Flexors are posterior in LL.

2. FUNCTIONAL DIFFERENCES

Upper Limb

- Prehension and grasping
- Fine motor activity

Lower Limb

- Locomotion
- Weight bearing
- Maintenance of posture and balance

3. GIRDLES

- UL is attached to the trunk by the **Pectoral (Shoulder) Girdle**.
- LL is attached to the trunk by the **Pelvic Girdle**.

II. COMPARISON OF BONES (COUNT)

Upper Limb	Count	Lower Limb	Count
Pectoral girdle	4	Pelvic girdle	2
Humerus	2	Femur	2
Radius & Ulna	4	Tibia & Fibula	4
Carpals	16	Tarsals	14
Metacarpals	10	Metatarsals	10
Phalanges	28	Phalanges	28
Total	64	Total	62

UPPER LIMB	LOWER LIMB
Prehension	Locomotion, Wt bearing
Smaller and weaker bones	Larger and strong
Joints are smaller and less stable	Larger and more stable
Muscles are smaller and attaches to smaller bony areas	Antigravity muscles more developed
Pectoral girdle is made of two bones	Pelvic girdle is made of single bone
Preaxial border faces laterally	Faces medially

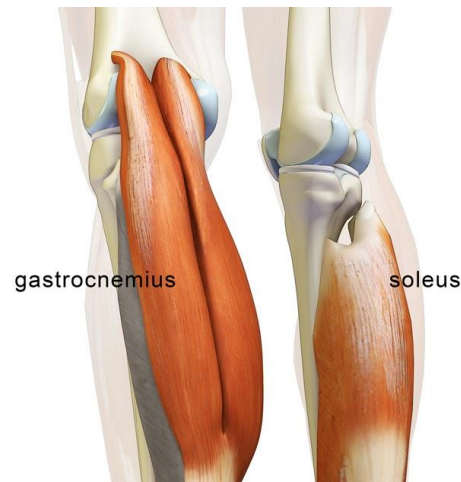
III. JOINTS, MUSCLES & VASCULAR FEATURES OF LL

JOINTS

- **Hip joint** – Very stable joint (deep acetabulum, strong ligaments).
- **Knee joint** – Largest joint of the body, but less stable.

MUSCULATURE

- LL muscles are **highly developed** because they:
 - Support body weight
 - Enable locomotion
- **Gastrocnemius and Soleus** act as the “peripheral heart”:
 - Help pump venous blood against gravity.



VASCULARITY

- Veins of the lower limb are well developed to **return blood against gravity**. There are 3 reasons behind this:

- ❖ Valve
- ❖ Muscles (Peripheral hearts) and
- ❖ Pressure from Artery itself (For deep veins)

IV. HAND VS FOOT

- The **hand** has strong grasping ability due to highly developed intrinsic muscles.
- **Thumb** contributes nearly **80% of hand function**.
- The **foot** is not designed for grasping; it is adapted for support and propulsion.

V. IMPORTANT TRANSITION AREAS (NEUROVASCULAR)

These are common sites of compression and clinical importance:

Upper Limb

- Axilla
- Cubital fossa
- Carpal tunnel

Lower Limb

- Popliteal fossa
- Tarsal tunnel

VI. UNIQUE FEATURES OF THE LOWER LIMB

- LL is the **longest region of the body**.

Longest structures in the body

- Longest vein – Great saphenous vein
- Longest artery – Femoral artery (continuing as anterior & posterior tibial arteries)
- Longest muscle – Sartorius
- Longest bone – Femur

VII. GLUTEUS MAXIMUS – FUNCTIONAL IMPORTANCE

- Powerful extensor of the hip.
- Assists in:
 - Standing from sitting position
 - Climbing stairs
 - Stepping into a vehicle
- Functionally linked with **latissimus dorsi** via thoracolumbar fascia.

VIII. DONOR SITES & CLINICAL RELEVANCE

BONE & TENDON GRAFTING

- **Fibula**

- Common bone graft because:
 - Does not bear weight
 - Does not contribute significantly to knee joint

- **Gracilis & Semitendinosus tendons**

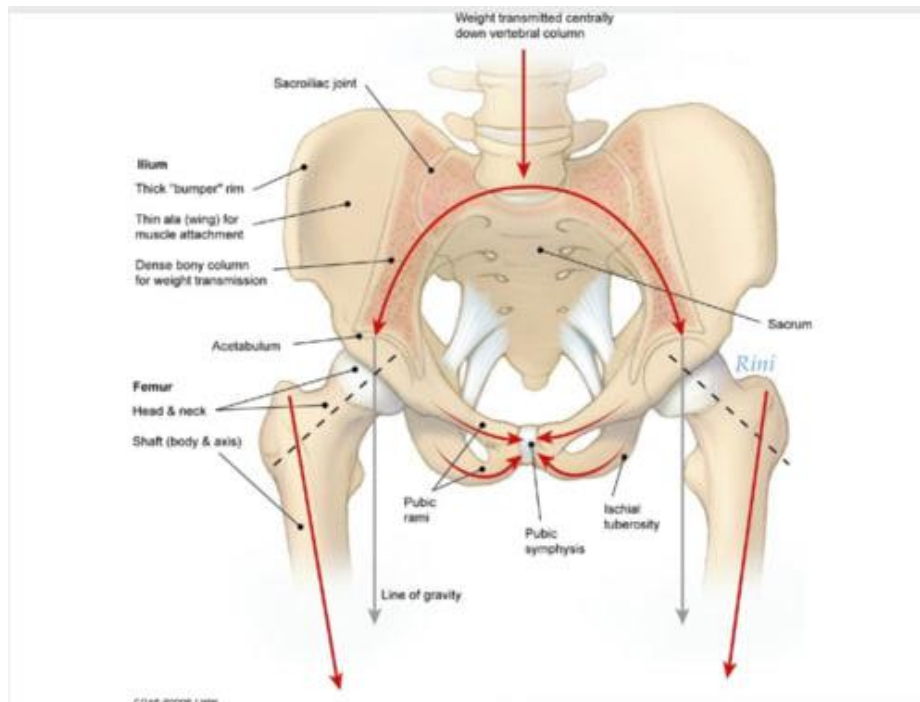
- Used in ligament reconstruction
 - Gracilis contributes less to adduction
 - Easily accessible
 - Muscles act synergistically, so loss is compensated by others

(Analogy: Palmaris longus in UL)

OTHER DONOR SITES

- Bone marrow aspiration – Sternum
- Vein graft – Great saphenous vein
 - Large size
 - Presence of alternative superficial veins
 - Fascia lata – Used in heart valve repair

IX. WEIGHT TRANSMISSION IN LOWER LIMB



Pathway:

1. Upper body → Right & Left pelvis
2. Pelvis → Head & neck of femur
3. Femur → Femoral condyles
4. Femoral condyles → Tibial condyles

• **Menisci** act as shock absorbers:

- Become compressed
 - Spread outward to distribute weight
- Weight is also transmitted anteriorly to the **pubic symphysis**.

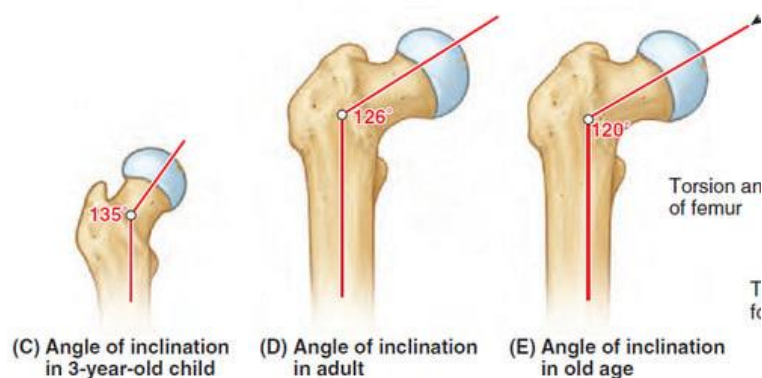
X. SITTING & SCIATIC NERVE

- Bone supporting body weight in sitting – **Ischial tuberosity**
- Pain during sitting is more common in males because:
 - Less subcutaneous fat
 - Wallet in back pocket compresses sciatic nerve
- Radiating pain down the limb indicates **sciatic nerve compression**.

XI. OBLIQUITY OF FEMUR & ANGLE OF INCLINATION

- Femur is **obliquely placed** because to bring body weight **toward the center of gravity**.

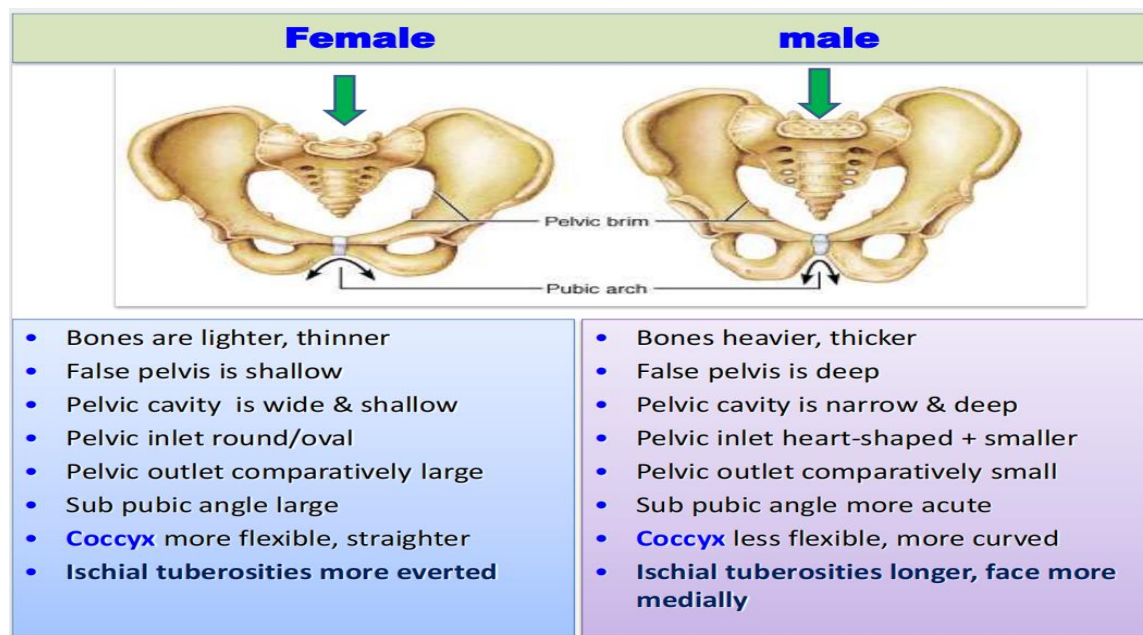
ANGLE OF INCLINATION (AOI)



- Normal adult: **123–128°**
- Inverse relationship with obliquity:
 - **↓ AoI → ↑ Obliquity**
(Inversely proportional!!!!)

SEX DIFFERENCES

- Females:
 - Wider pelvis (because of what? Childbearing!)
 - Acetabulum placed more laterally
 - Smaller AoI
 - Higher femoral obliquity
- Purpose: Balance body weight during upright posture.



XII. PELVIC GIRDLE

COMPONENTS (3 BONES)

1. **Ilium** – Upper part
2. **Pubis** – Lower & anterior
 - Body
 - Superior ramus
 - Inferior ramus

3. Ischium – Lower & posterior

- Body
- Tuberosity
- Ramus

XIII. ILIUM – ANATOMY

OUTER SURFACE

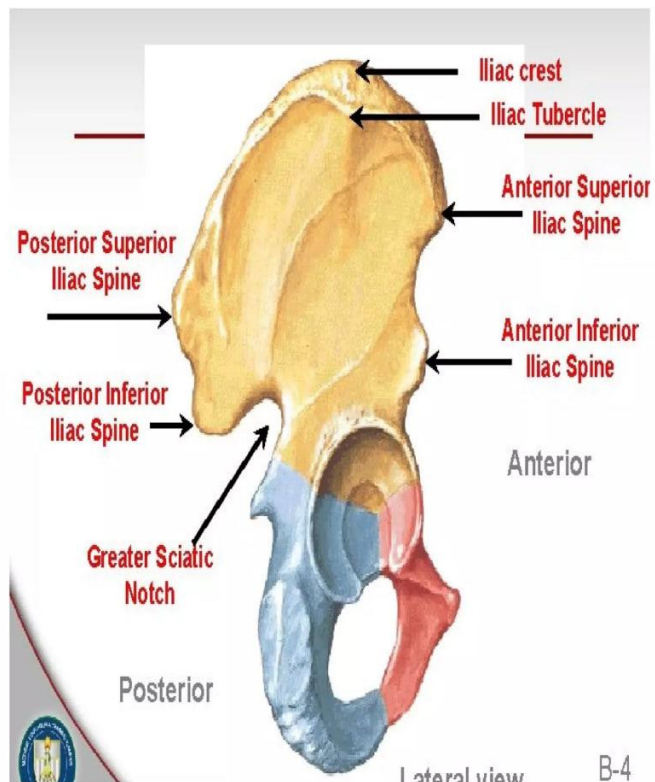
Ilium

Upper expanded plate like part of the hip bone.

It's lower part forms the upper 2/5 of acetabulum

The ilium has the following

1. **Upper end** – iliac crest
2. **Lower end** – which fused with pubis and ischium at the acetabulum
3. **Three borders** (anterior, posterior, medial)
4. **Three surfaces** (gluteal surface, iliac surface or iliac fossa and sacropelvic surface)

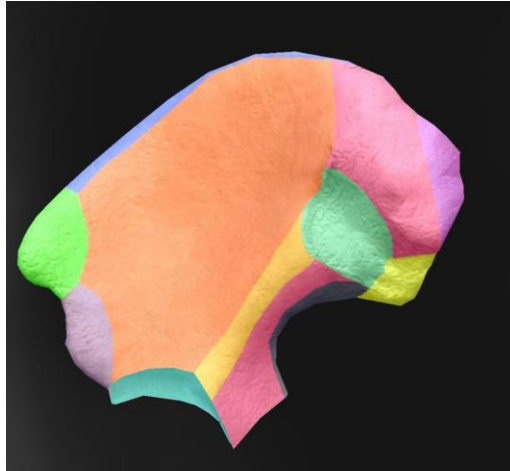


- Iliac crest (superior border)
- ASIS & AIIS (anterior border)
- PSIS & PIIS (posterior border)
- Greater sciatic notch (below PIIS)

- **Gluteal lines:**

- Posterior → Gluteus maximus
- Middle → Gluteus medius
- Inferior → Gluteus minimus

INNER SURFACE



- Arcuate line

- Above → Iliac fossa
- Below → Pelvic surface
 - Iliac tuberosity
 - Auricular surface (articulates with sacrum)

XIV. ISCHIUM & PUBIS: NORMAL NW!

XV. ORIENTATION POINTS OF HIP BONE

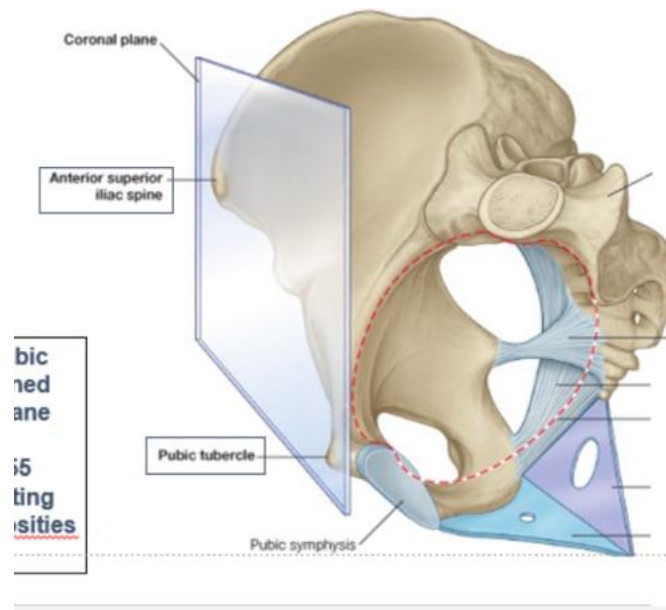
- Iliac crest – Upward
- Acetabulum – Lateral
- Pubis – Anterior

- Pubic crest:
 - Forms anterior border of pubic symphysis
 - Ends at pubic tubercle
 - Provides attachment for abdominal muscles

XVI. ANATOMICAL POSITION OF THE HIP BONE

The hip bone is said to be in anatomical position when the following criteria are satisfied:

- The **Anterior Superior Iliac Spine (ASIS)** and the **anterosuperior aspect of the pubic symphysis** lie in the same **coronal plane**.



- The **acetabulum faces inferolaterally**.
- The **acetabular notch** is directed **inferomedially**.
- The **obturator foramen** is oriented inferiorly and slightly anteriorly, **aligned beneath the acetabulum**.
- The **internal surface of the body of the pubis** faces almost directly **superiorly**.

XVII. JOINTS OF THE PELVIC GIRDLE

- **Posterolateral joint → Sacroiliac joint**

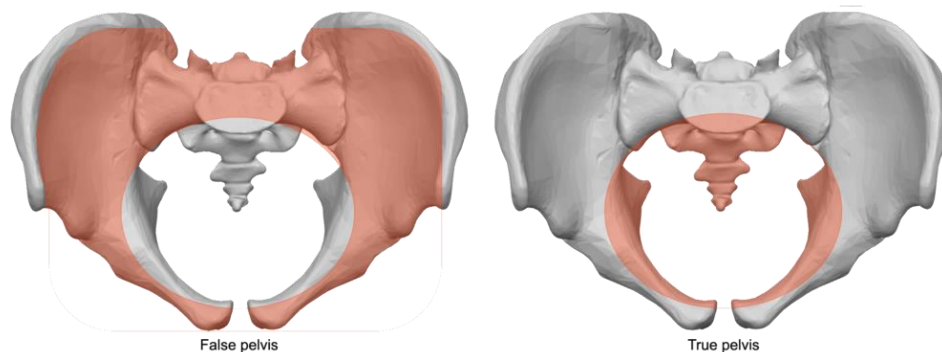
- Strong synovial joint (Infact it is a mix of fibrocartilage and synovial (fluid-filled) parts)
- Transmits body weight from vertebral column to lower limbs

- **Anteromedial joint → Pubic symphysis**

- A **secondary cartilaginous joint (complex symphysis)**
- Reinforced by ligaments and fibrocartilage
- Supported anteriorly by superficial fascia and subcutaneous fat of the pubic region

XVIII. PELVIC INLET (TRUE PELVIC BRIM)

- Forms the **junction between abdomen and pelvis.**
- Divides the bony pelvis into:
 1. **Greater (False) Pelvis** – above pelvic inlet
 2. **Lesser (True) Pelvis** – below pelvic inlet



DIVISIONS OF THE PELVIS

- **Greater (False) Pelvis** – contains **inferior abdominal viscera** (e.g., loops of ileum, sigmoid colon, appendix may descend here).
- **Lesser (True) Pelvis** – contains **pelvic viscera** (urinary bladder, rectum, internal reproductive organs).

INFERIOR ABDOMINAL / ILIAC FOSSA STRUCTURES

- **Iliac fossa muscle: Iliacus muscle**
 - Origin: Iliac fossa
 - Insertion: Lesser trochanter of femur (via iliopsoas tendon)
 - Function: **Chief flexor of the hip joint**
- **Clinical correlation – Psoas sign**
 - In inflammation of the appendix, contraction of iliopsoas causes pain during hip flexion
 - Inability or pain while flexing the hip indicates **Positive Psoas Sign**
 - Caused by irritation/compression of the muscle by inflamed appendix

PELVIC NERVES – MAJOR EXITS ("WINDOWS")

There are **two major openings** from the pelvis:

GREATER SCIATIC FORAMEN

- Opens from pelvis to **gluteal region**
- **Piriformis muscle** passes through and **divides it into two parts**

ABOVE PIRIFORMIS

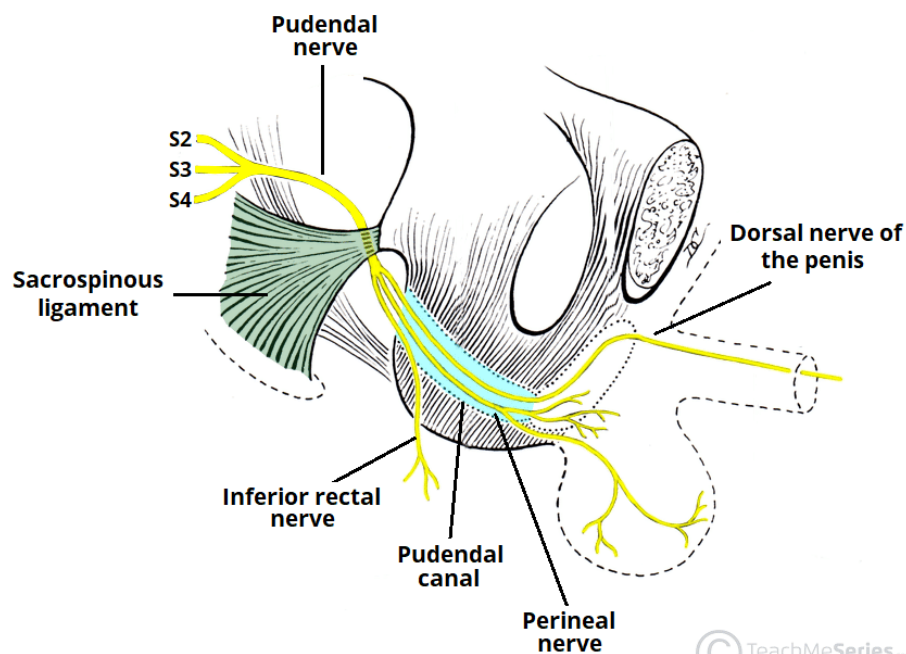
- Superior gluteal nerve, artery & vein

BELOW PIRIFORMIS

- Sciatic nerve (L4–S3)
- Inferior gluteal nerve, artery & vein
- Posterior femoral cutaneous nerve
- Pudendal nerve
- Internal pudendal artery & vein
- Nerve to obturator internus & superior gemellus
- Nerve to quadratus femoris & inferior gemellus

LESSER SCIATIC FORAMEN

- Tendon of obturator internus muscle
- Pudendal nerve
- Internal pudendal artery & vein
- **Function:** Pudendal nerve passes from **greater sciatic foramen** → **lesser sciatic foramen** → **perineum**



OBTURATOR CANAL

- Contents:
 - Obturator nerve
 - Obturator artery & vein
- Supplies **medial compartment of thigh**
 - Motor: Adductor muscles
 - Sensory: Medial thigh skin
- **Clinical note:**
 - Loss of sensation over medial thigh → suspect **Obturator nerve injury**

NERVE INJURY ASSESSMENT (CLINICAL APPROACH)

- Start with **sensory examination**
- Follow with **motor assessment** (adductor muscles)
- Identify the level of injury by **retrograde examination** (distal → proximal)
- First assess **obturator canal** (tumor/compression)
- Then evaluate **spinal roots** (L2, L3, L4)
- Compare with **femoral nerve**, as both share same root values but different pathways

RETRO-INGUINAL SPACE (SUBINGUINAL / BORGIO'S SPACE)

Divided into **two compartments**:

1. LATERAL COMPARTMENT – LACUNA MUSCULORUM (NEUROMUSCULAR)

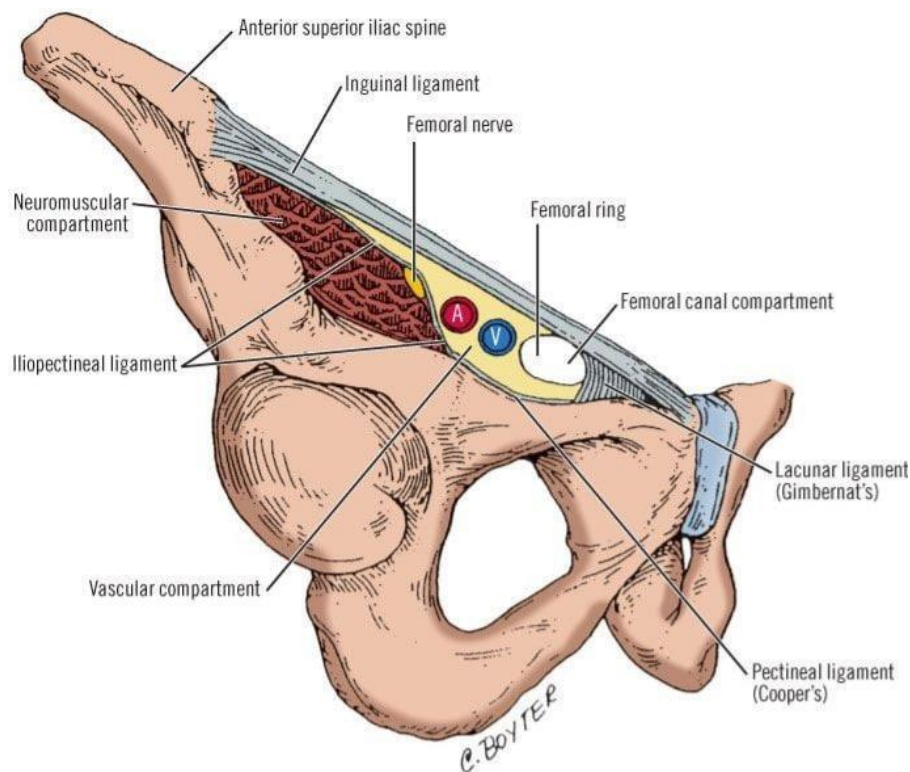
- Psoas major muscle
 - Iliacus muscle
 - Femoral nerve

2. MEDIAL COMPARTMENT – LACUNA VASORUM

- Femoral artery
- Femoral vein
- Femoral branch of genitofemoral nerve
- Lymphatics

• Related ligaments:

- Inguinal ligament
- Lacunar ligament
- Pectineal (Cooper's) ligament



INGUINAL LIGAMENT

- Forms boundary between **anterolateral abdominal wall and thigh**
- Important surgical and anatomical landmark

ILIAC CREST

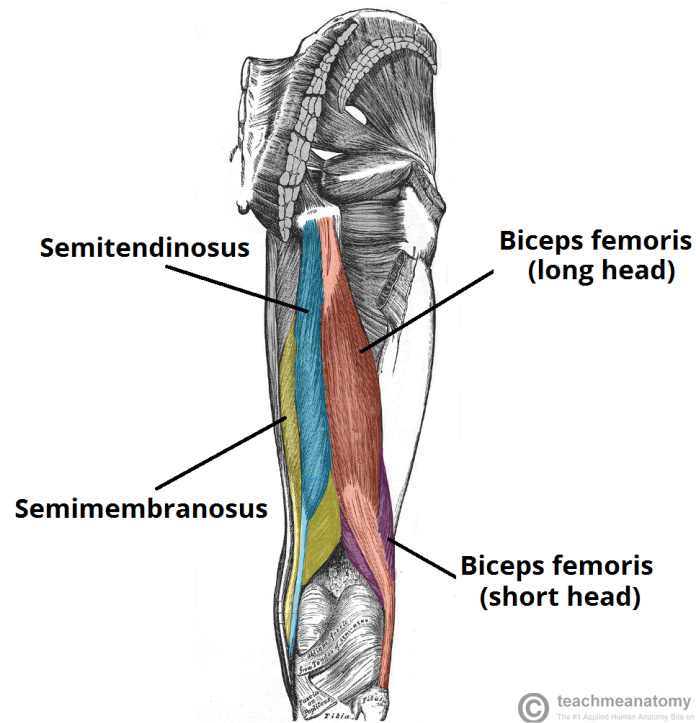
- Easily **palpable landmark**
- Used to:
 - Identify pelvic level
 - Locate L4 vertebral level (line joining highest points of iliac crests)
- Helpful for procedures such as **lumbar puncture**

TRIRADIATE CARTILAGE

- Cartilage in the acetabulum between ilium, ischium, and pubis
- Fuses completely at **~25 – 28 years of age**

ISCHIAL TUBEROSITY & HAMSTRINGS

- Hamstring muscles originate from ischial tuberosity
- Sudden hip flexion/extension → risk of hamstring injury due to stress at origin



SEXUAL DIMORPHISM OF PELVIS

MALE PELVIS

- Pelvic inlet: **Heart-shaped**
- Subpubic angle: **< 90°**
- Narrower pelvic cavity and outlet

FEMALE PELVIS

- Pelvic inlet: **Oval / round**
- Subpubic angle: **> 90°**
- Wider pelvic cavity and outlet (adapted for childbirth)

PELVIC DIAMETERS

PELVIC INLET

- Transverse diameter: ~13 cm
- Oblique diameter: ~12 cm
- Anteroposterior diameter: ~11 cm

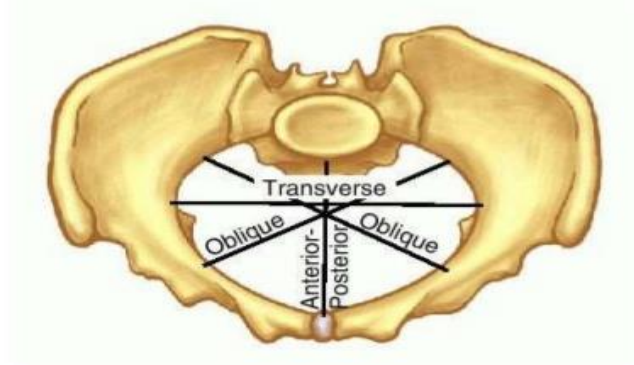
PELVIC OUTLET

- Boundaries:
 - Posterior: Coccyx
 - Anterior: Inferior border of pubic symphysis
- Anteroposterior diameter: ~13 cm
- Oblique diameter: ~12 cm
- Transverse diameter: ~11 cm

PELVIC CAVITY

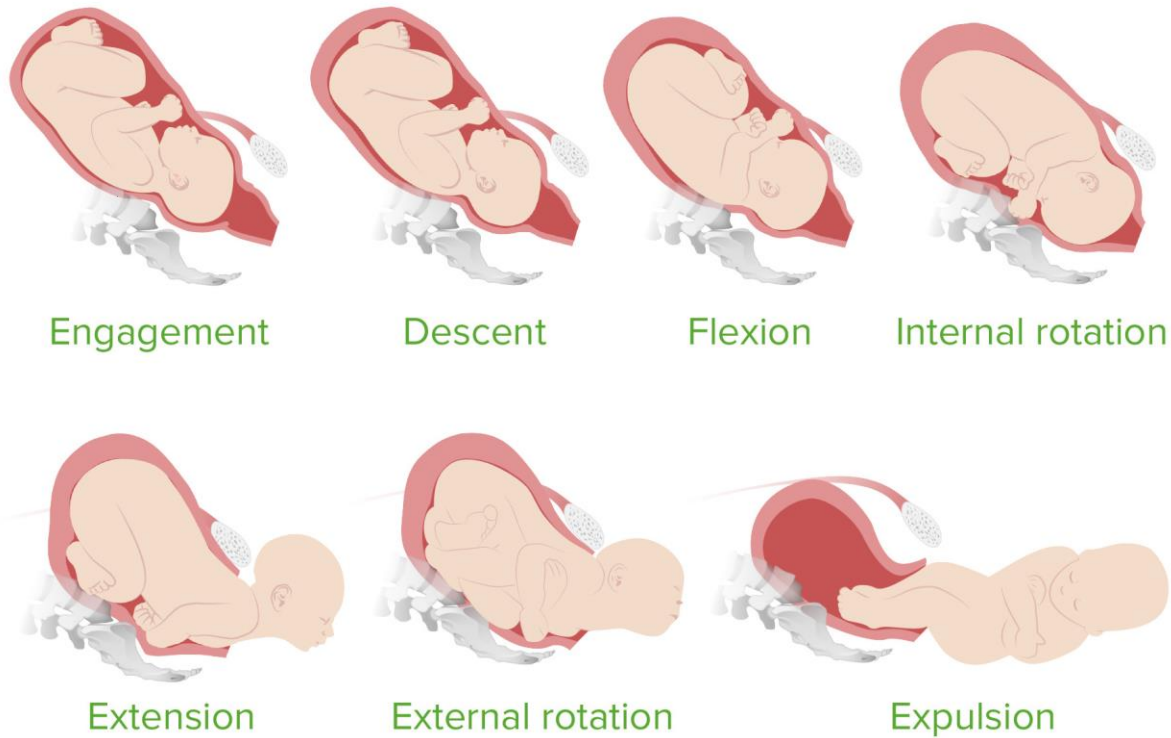
- Diameters are approximately **equal (~12 cm)**

Pelvis	Diameter (cm)		
	Anteroposterior	Transverse	Oblique
Pelvic Inlet	11	13	12
Pelvic Cavity	12	12	12
Pelvic Outlet	13	11	-



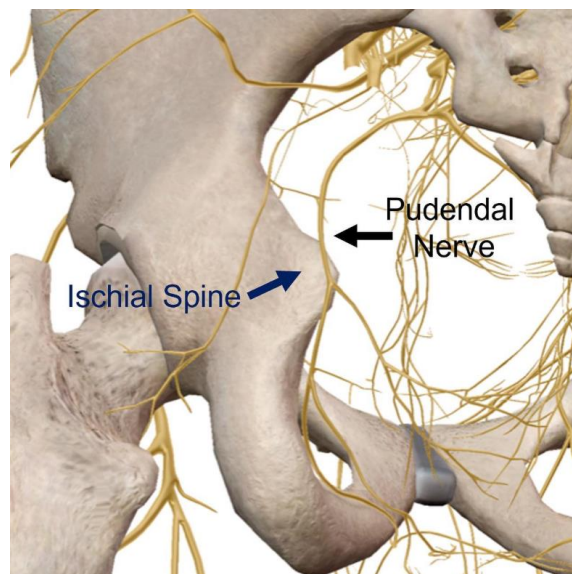
OBSTETRIC & ANESTHETIC CORRELATIONS

- Fetal head (Large Anterio-posterior diameter length) enters pelvis in **transverse diameter** (widest)
- During labor, head rotates to **anteroposterior diameter** at outlet
- Final position: fetal head faces **anteriorly**



- **Ischial spines:**

- Important obstetric landmarks
- Pudendal nerve passes nearby (branch of sacral plexus)



- **Sacral plexus** exits pelvis via greater sciatic foramen

CSF & LUMBAR PUNCTURE

- Adults: Below **L1–L2** level
- Children: **L3–L4** level
- Prevents spinal cord injury

PUDENDAL NERVE BLOCK (LABOR ANALGESIA)

- Local anesthetic injected near **ischial spine / coccyx region**
- Used for pain relief during labor

4 P'S OF LUMBAR PUNCTURE / ANESTHESIA

1. Preparation of the patient
2. Protection of local structures
3. Proper positioning (anatomical)
4. Precise puncture technique

Ara Mohammed